23. Algebraic and Differential Invariants

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If I am to report on the development of algebraic and differential invariants, then in both areas I should like to restrict myself to the critical period which, according to a remark of Hilbert, follows the naive and formal period. This critical period is characterized, for the algebraic invariants, by the name of Hilbert himself; for the differential invariants, by the name of Riemann — or, in substantive terms: for the algebraic invariants by the arithmetical methods of algebra, which essentially developed around these questions and were able here to show their full sharpness; for the differential invariants by the methods of the formal calculus of variations. Thus I should like to report on the arithmetical methods with their significance beyond the special subject, while in the second part I can restrict myself to the subordination of the differential invariants to the algebraic ones, which is accomplished precisely in connection with the methods of the calculus of variations.

1. The *algebraic invariants* are, as is well known, based on the *general linear group* in $x_1, \ldots, x_n$:

$$x_i = \sum_k s_{ik} x'_k \qquad \text{or briefly:} \qquad x = S(x'), \tag{1}$$

where the s_{ik} denote indeterminates, so that $\Delta = |s_{ik}|$ is non-zero. If now $f(a, x)$, $g(b, x), \ldots$ are forms in x, of dimensions $\alpha, \beta, \ldots$, with indeterminates $a, b, \ldots$ as coefficients, then by means of (1) the *induced transformation* of the $a, b, \ldots$ arises; for from

$$\begin{aligned} f(a,x) &= \sum a_{i_1 \ldots i_n} x_1^{i_1} \cdots x_n^{i_n} = \sum a'_{j_1 \ldots j_n} {x'_1}^{j_1} \cdots {x'_n}^{j_n} = f(a', x'), \\ g(b,x) &= \sum b_{r_1 \ldots r_n} x_1^{r_1} \cdots x_n^{r_n} = \sum b'_{s_1 \ldots s_n} {x'_1}^{s_1} \cdots {x'_n}^{s_n} = g(b', x') \end{aligned}$$

it follows that the $a', b', \ldots$ become linear homogeneous functions of the $a, b, \ldots$, whose coefficients are of degrees $\alpha, \beta, \ldots$ in the s_{ik}:

$$a' = A_\alpha(a); \qquad b' = B_\beta(b) \ldots. \tag{2}$$

By an *invariant* one now understands an invariant with respect to the transformations (1) and (2); that is, an integral rational function of the indeterminates $a, b, \ldots, x, y, \ldots$ — where also $y = S(y')$ — which, under application of the transformations, reproduces itself up to a factor, a power of the substitution determinant:

$$I(a', b', \ldots, x', y', \ldots) = \Delta^\rho I(a, b, \ldots, x, y, \ldots). \tag{3}$$

The coefficients of I may be assumed to be rational, or even rational integers, since every invariant with coefficients in an arbitrary number field P can be expressed linearly, with coefficients from P, by finitely many such special ones. Likewise it is no restriction to assume I separately homogeneous in each series, since here again every invariant can be assembled as a sum of finitely many such special ones.

It should be noted that conversely (1) and (2) are again determined by the requirement (3). As Ostrowski and Schur have recently shown,[1] the induced transformations can also be characterized as the totality of the linear transformations, separated in each individual series, which — apart from certain specifiable exceptions — allow an arbitrary invariant I free of the $x, y, \ldots$.

The product of two invariants is again an invariant, but the sum is one if and only if the weight — the exponent ρ in (3) — agrees. If, however, one restricts oneself to transformations of determinant one, then any sum is again an invariant and also exhausts all invariants with respect to the so restricted group. One therefore obtains an integral domain, indeed a *homogeneous integral domain*, that is, a domain for which membership of a polynomial implies membership of its homogeneous components separately — which here again amounts to the separately homogeneous invariants according to (3).

Here arises the fundamental problem around which invariant theory, and arithmetical algebra in general, has developed: *Is this integral domain finite*, that is, does it possess a *finite integral basis* $I_1, \ldots, I_k$ such that every invariant can be represented integrally and rationally through $I_1, \ldots, I_k$, $I = G(I_1, \ldots, I_k)$? Hilbert's solution of the problem — Gordan's proof for the binary case does not admit an extension because of its unmanageable symbolic computations, and the Mertens–Hilbert proof does not do so because of its use of the decomposition of a binary form into linear forms — rests on reducing the question of the integral basis to that of the ideal basis. I should like to sketch the train of thought here, emphasizing the arithmetical

[1] A. Ostrowski and I. Schur, Über eine fundamentale Eigenschaft der Invarianten einer binären Form, Math. Zeitschr. 15 (1922), and related, not yet published work of Ostrowski.

fundamental ideas, and then to go on to three connected circles of questions: actual construction of the basis by finitely many steps; finiteness questions for subgroups; finiteness questions taking integrality into account.

2. I recall the *definition of an ideal* for finite algebraic number fields. A system $\mathfrak{a}$ of numbers of the field is called an ideal if

1. $\mathfrak{a}$ belongs to the domain $\mathfrak{o}$ of all integers of the field,
2. together with α and β, the difference $\alpha - \beta$ also belongs to $\mathfrak{a}$,
3. together with α, $\lambda\alpha$ also belongs to $\mathfrak{a}$, where λ is any element of $\mathfrak{o}$.

And, as is well known, the theorem holds that every ideal has an *ideal basis*: $\mathfrak{a} = (\alpha_1, \ldots, \alpha_k)$; that is, there are finitely many elements $\alpha_1, \ldots, \alpha_k$ from $\mathfrak{a}$ such that, by means of 2. and 3., $\mathfrak{a}$ is derivable from $\alpha_1, \ldots, \alpha_k$; thus $\alpha = \lambda_1\alpha_1 + \cdots + \lambda_k\alpha_k$ exhausts all elements from $\mathfrak{a}$.[2]

This definition of an ideal rests only on the fact that $\mathfrak{o}$ forms an integral domain; the concept of ideal, and likewise that of ideal basis, therefore remains literally the same when $\mathfrak{o}$ is replaced by an arbitrary integral domain.

Hilbert's finiteness proof now decomposes into the following three theorems:

1. In the domain of all polynomials in n indeterminates with coefficients in a field of rationality, the ideal-basis theorem holds for every ideal — and consequently also for every arbitrary system $\mathfrak{S}$.
2. A homogeneous integral domain $\mathfrak{J}$ of polynomials is finite if and only if the ideal-basis theorem holds in $\mathfrak{J}$.
3. For the domain of invariants $\mathfrak{J}$, the ideal-basis theorem holds in the sharpened form that every ideal basis formed with respect to the domain of all polynomials is at the same time an ideal basis in $\mathfrak{J}$, so that beside

$$I = A_1 I_1 + \cdots + A_k I_k \qquad \text{one always has} \qquad I = i_1 I_1 + \cdots + i_k I_k,$$

where the A denote polynomials in the $a, b, \ldots, x, y, \ldots$, and the i denote invariants.

The proof of 1. rests, in principle, on the same arguments as in the algebraic number field; in both cases the existence of the ideal basis follows directly from a general theorem on modules of linear forms.[3] From 1. the existence of the ideal basis follows directly for every finite integral domain of polynomials; for homogeneous domains, the converse is also easy to see. Only in 3. are special properties of invariants used; namely, in Hilbert, 3. follows from a known theorem on the Ω-process, whereas recently E. Fischer has shown that 3. can — on the basis of other general theorems — already be obtained from the property of the general linear group that, along with every transformation, it also contains the contragredient one, respectively its conjugate-complex one.[4]

3. The *actual construction of the basis* rests on a further arithmetical transformation of the notion of finiteness by drawing on the theory of *function fields.* The following holds:

4. An integral domain $\mathfrak{J}$ of polynomials is finite if and only if there is contained in $\mathfrak{J}$ a finite subdomain $\mathfrak{J}'$ such that $\mathfrak{J}$ depends algebraically integrally on $\mathfrak{J}'$; every integral basis of $\mathfrak{J}'$ is then supplemented by a fundamental system of these algebraically integral quantities to a basis of $\mathfrak{J}$.

[5] If, specifically, as in the case of the invariants, one deals with homogeneous domains $\mathfrak{J}$ for which the ideal-basis theorem holds in the sharper form 3., then such a domain $\mathfrak{J}'$ is derivable from some always-existing finitely many polynomials $I_1, \ldots, I_s$ as an integral basis, from whose vanishing the vanishing of all polynomials from $\mathfrak{J}$ follows. This fact is again based on a general theorem of ideal theory:

5. To every ideal $\mathfrak{a}$ of polynomials there belongs an ordinary rational integer r such that for every ideal $\mathfrak{b}$ which vanishes at all zeros of $\mathfrak{a}$, $\mathfrak{b}^r$ is divisible by $\mathfrak{a}$.

[2] That k can be reduced to two is irrelevant for what follows.

[3] Compare my comparative report on the arithmetical theory of algebraic functions. Jahresber. 28 (1920), p. 187 and note 2), p. 188.

[4] E. Fischer, Über die Endlichkeit der Invarianten. Göttinger Nachrichten 1915, and Leipzig lecture.

[5] Hilbert shows (Über die vollen Invariantensysteme, Math. Annalen 42 (1893), §§ 1 and 2) that these facts follow from the assumed finiteness; conversely, that finiteness follows from algebraically integral dependence follows from the existence of the rational basis (E. Noether, Körper und Systeme rationaler Funktionen, Math. Annalen 76 (1915), § 12).

In the case of invariants, one can determine an *upper bound for the degrees* of $I_1, \dots, I_s$, and thus also for their number s, depending only on the degrees of the basic forms under consideration. At this point, again, special properties of invariants are used; the relevant theorem says that a numerically specialized system of basic forms has a non-zero invariant if and only if $\Delta = |s_{ik}|$ depends algebraically integrally on the transformed coefficients.

From this upper bound K. Hentzelt[6] has determined an *upper bound for the degrees of the invariants of the integral basis*, again depending only on the degrees of the basic forms. He succeeds in doing this by giving, for the exponent r in 5., an upper bound depending only on the number and the highest degree of the polynomials of an ideal basis of $\mathfrak{a}$, independent of the coefficients of these polynomials, and computable from the given numbers in finitely many steps. In principle the problem posed is thereby completely settled; admittedly the bound given is quite high. Thus, for a ternary quadratic form, where the discriminant — that is, an invariant of degree 3 — already forms the basis of all invariants free of the x, according to Hilbert and Hentzelt one reaches high into the millions.

4. In the finiteness question for invariants of *subgroups* of the general linear group, only partial results have so far been reached. The method rests almost exclusively on reducing the invariants of subgroups to simultaneous invariants of the general linear group, according to the procedure used by Study for the orthogonal group. This has been carried out by Weitzenböck for motion invariants and affine invariants, and by Deruyts for semi-invariants and shear invariants;[7] the question of the reach of this method is still undecided. Fischer's remark mentioned at the end of 2. likewise solves the finiteness problem for a series of subgroups; in particular, it thereby gives the simplest proof for the orthogonal group. But the example of semi-invariants shows — as Fischer has recently pointed out[8] — that for subgroups, despite finiteness, the sharper form 3. for the ideal basis need not be fulfilled.

Interpreted in the sense of field theory, the finiteness question present here leads to Hilbert's problem of *relatively integral functions*, that is, to the question whether such integral domains as consist of the totality of the polynomials of a field of rational functions are always finite. In this direction lies the elementary provable fact that for the invariants of finite groups a distinguished integral basis can be given directly — the system of coefficients of the Galois resolvent.[9]

5. The question whether the finiteness theorem for invariants can be sharpened with respect to *integrality* is also not yet settled in general. To be sure, as Hilbert showed, the ideal-basis theorem also holds for the domain of all integral polynomials; and likewise the connection 2. between ideal basis and integral basis remains valid;[10] but the proof for the — unnecessary — sharpened version 3. fails. For the case of binary invariants, integral finiteness can be proved;[11] the proof, which represents an integrality sharpening of the binary Mertens–Hilbert finiteness proof, uses, besides the theorem on the integral ideal basis, also the decomposition of binary forms into linear forms, and therefore does not permit transfer to more indeterminates. On similar foundations, by contrast, integral finiteness for the invariants of finite groups follows, as a sharpening of the proof mentioned above, although now again only an existence proof and no actual specification of the basis results.[12] Here too, only a further development of the theory of function fields and of general ideal theory will presumably lead to the goal.

6. I now pass to the *differential invariants* and to the question raised in the introduction concerning their reduction to the theory of algebraic invariants. The underlying group is, as is well known, defined by means of $x_i = x_i(y)$ as:

$$\mathrm{d}x_i = \sum_k \frac{\partial x_i}{\partial y_k}\mathrm{d}y_k; \qquad \mathrm{d}\delta x_i = \sum_k \frac{\partial x_i}{\partial y_k}\mathrm{d}\delta y_k + \sum_{k,l} \frac{\partial^2 x_i}{\partial y_k \partial y_l}\mathrm{d}y_k\,\delta y_l;\ \dots. \tag{4}$$

To pass to the transformations induced by (4), one may begin with an arbitrary function $f(x, \mathrm{d}x) = \varphi(y, \mathrm{d}y)$; by requiring the invariance of $\delta f, \delta^2 f, \dots$ with respect to (4), one obtains the transformations for the derivatives of f with respect to x and $\mathrm{d}x$. If one restricts oneself, for instance, to forms homogeneous in the $\mathrm{d}x$:

$$f(x, \mathrm{d}x) = \sum a_{i_1 \dots i_n}(x)\mathrm{d}x_1^{i_1} \cdots \mathrm{d}x_n^{i_n} = \sum a'_{i_1 \dots i_n}(y)\mathrm{d}y_1^{i_1} \cdots \mathrm{d}y_n^{i_n} = \varphi(y, \mathrm{d}y),$$

[6] I shall shortly publish the work from the estate.

[7] For literature references, compare Weitzenböck's encyclopedia article on invariant theory; III E 1.

[8] Leipzig lecture and Math. Zeitschrift.

[9] E. Noether, Der Endlichkeitssatz der Invarianten endlicher Gruppen. Math. Ann. 77 (1916).

[10] From this connection it follows in particular that the decomposition theorems of general ideal theory hold for all finite integral domains, as I developed them in Math. Ann. 83 (1921).

[11] E. Noether, Die Endlichkeit des Systems der ganzzahligen Invarianten binärer Formen. Göttinger Nachrichten 1919.

[12] § 4 of the work cited under 11).

then the a' are linear in the a, the $\frac{\partial a'}{\partial y}$ are linear in the a and $\frac{\partial a}{\partial x}$, and so on:

$$a' = A_0(a), \qquad \frac{\partial a'}{\partial y} = A_1\left(a, \frac{\partial a}{\partial x}\right), \qquad \frac{\partial^2 a'}{\partial y^2} = A_2\left(a, \frac{\partial a}{\partial x}, \frac{\partial^2 a}{\partial x^2}\right), \ldots \tag{5}$$

and the question is one of invariance with respect to the transformations (4) and (5). Here all the quantities occurring,

$$a, \frac{\partial a}{\partial x}, \ldots, \mathrm{d}x, \delta x, \mathrm{d}\delta x, \ldots, \frac{\partial x}{\partial y}, \frac{\partial^2 x}{\partial y^2}, \ldots,$$

are to be regarded as indeterminates, so the determinant of each individual transformation is certainly non-zero. Only when the formally completed theory is applied to special functions does the question arise of restriction to such functions that the transformations remain uniquely reversible in a certain domain and that a sufficient number of differential quotients exist, exactly as in the algebraic case, under numerical specialization, the determinant $|s_{ik}|$ must be assumed non-zero.

Under this treatment of all arguments occurring as indeterminates, one is thus dealing with a special linear group, not directly accessible to the ordinary algebraic methods, in which the transformations of $\mathrm{d}x$ and $a' = A_0(a)$ agree with (1) and the induced (2). The question arises whether, perhaps quite generally, by the *introduction of new arguments*, the transformations (4) and (5) can be reduced to the general linear group (1) and the transformations (2) thereby *induced*.

This is in fact the case. The higher differentials $\mathrm{d}^2x, \delta\mathrm{d}x, \ldots$ are to be replaced by the Lagrange derivatives arising from the variational problem belonging to $f(x, \mathrm{d}x)$, and by further combinations formed from them by polar formation (transmissions in the sense of Weyl and Schouten), all of which are cogredient to the $\mathrm{d}x$, so that their transformation is given by the general linear group (1). The quantities $\frac{\partial a}{\partial x}, \frac{\partial^2 a}{\partial x^2}, \ldots$ (or, more generally, the derivatives of the homogeneously assumed f with respect to x and $\mathrm{d}x$), however, are to be replaced by the coefficients of certain forms arising from the "normal forms of the higher variations" by the above elimination of the $\mathrm{d}^2x, \delta\mathrm{d}x, \ldots$, and of their "covariant derivatives"

$$[\Omega_i], \quad [\Omega_i^{(1)}], \quad [\Omega_i^{(2)}], \ldots \qquad (i = 1, 2, \ldots);$$

and since the $[\Omega_i^{(k)}]$ are invariants with respect to (4) and (5) which depend only on the $\mathrm{d}x$ and δx, these coefficients satisfy the algebraically induced transformations (2).[13] Through this *reduction theorem*, the theory of differential invariants is subordinated to algebraic invariant theory.

The series of the $[\Omega_i]$ breaks off with $[\Omega_\alpha]$ if $f(x, \mathrm{d}x)$ is a homogeneous form of α-th dimension in the $\mathrm{d}x$. For quadratic forms, $[\Omega_2]$ becomes identical with Riemann's curvature form, and the formation process indicated here is exactly the one given by Riemann in the Latin prize essay, which Lipschitz, independently of this and in connection with Riemann's habilitation lecture, also developed.[14] The proof of the reduction theorem, which Christoffel and Ricci had carried out computationally for quadratic differential forms, rests on the introduction of "Riemann normal coordinates," which transform the extremals emanating from a point into straight lines; since thereby to an arbitrary transformation of the x there corresponds a linear transformation of the normal coordinates, these determine the passage to the general linear group.

The question arises whether such a reduction theorem also exists when, following Weyl and Schouten, the group is defined not by a variational problem but only by a transmission; that is, when, corresponding to the Lagrange derivatives, expressions $\psi_i(\mathrm{d}\delta x, \mathrm{d}x, \delta x)$ are set up through which the elimination of the higher differentials is achieved, with the restriction, unnecessary in itself, that the ψ_i should be linear in the $\mathrm{d}x$ being added in analogy with quadratic differential forms (for arbitrary homogeneous $f(x, \mathrm{d}x)$, the polarized Lagrange derivatives become homogeneous of first order only in $\mathrm{d}x$). By the requirement of cogredience of ψ to the $\mathrm{d}x$, the transformation of their coefficients is induced, and thus, in analogy with (5), the group is fixed. For invariants of the lowest orders, Weitzenböck has confirmed the reduction theorem computationally in the Weyl case; a general Ansatz is still lacking.[15]

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[13] E. Noether, Invarianten beliebiger Differentialausdrücke. Göttinger Nachrichten 1918.

[14] Compare also my account of Riemann and Lipschitz in no. 28 of the second part of Weitzenböck's encyclopedia article mentioned under 7).

[15] Compare: Weyl, Raum, Zeit, Materie; Schouten, Math. Zeitschr. 13 (1922); Weitzenböck, Sitzungsber. d. Wiener Akademie, 1920 and 1921.